MICROPROCESSOR AND MICROCONTROLLERS

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Exp 01: Analysis Development and Verification of 8 bit and 16 bit Arithmetic Operations using Assembly Language

Objectives:

1. To Analysis and Verify 8 bit Addition and Subtraction using 8 bit Micro Processor
2. To Analysis and Verify 16 bit Addition and Subtraction using 8 bit Micro Processor
3. To Analysis and Verify 8 bit Multiplication and Division using 8 bit Micro Processor
4. To Analysis and Verify 16 bit Multiplication and Division using 8 bit Micro Processor

Components Required:

8085 Simulator

Procedure:

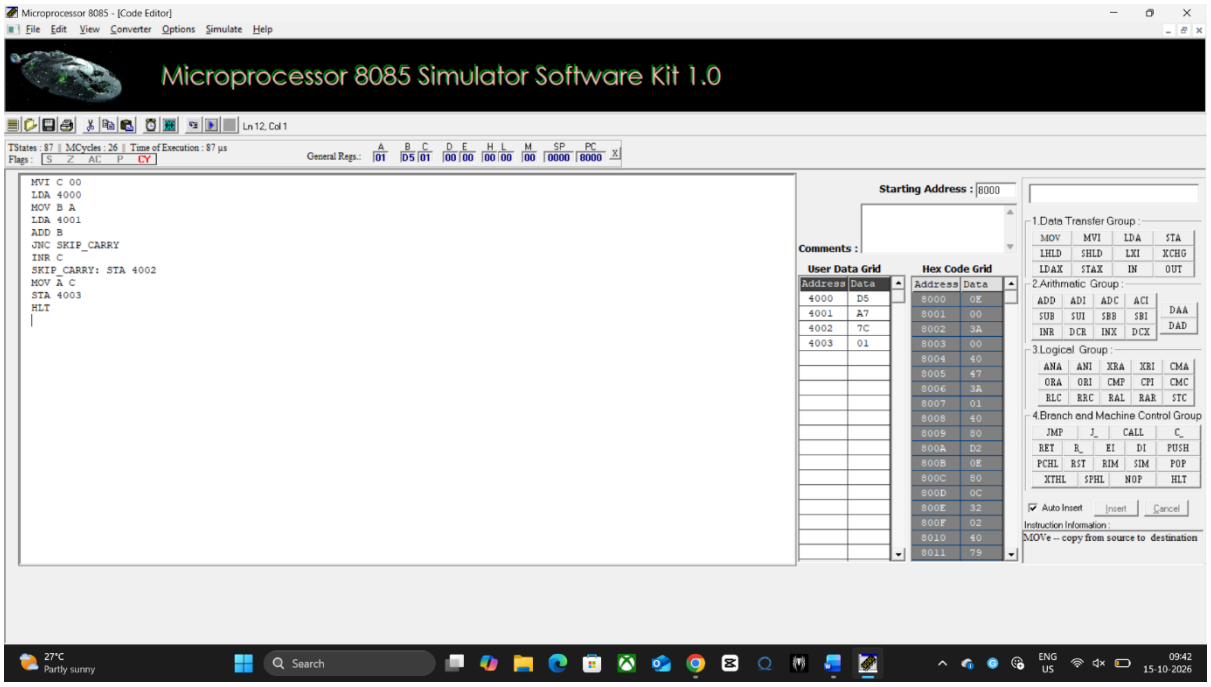
1. Launch your 8085 Simulator application on your system.
2. Type the Assembly Language Program (ALP) into the editor/assembler window.
3. Click on the Assemble or Compile button to convert the assembly mnemonics into machine code. Ensure there are no syntax errors.
4. Open the Memory Editor window and manually enter your input hex operands into the designated starting memory addresses.
5. Click Run or use Step-by-Step execution to trace the accumulation and flag changes.
6. Check the specified target memory locations and register statuses to verify your mathematical outputs.

1. To Analysis and Verify 8 bit Addition and Subtraction using 8 bit Micro Processor

Code:

Addition:

```
LDA 4000
MOV B A
LDA 4001
MVI C 00
ADD B
JNC SKIP
  INR C
SKIP:
  STA 4002
  MOV A C
  STA 4003
  HLT
```

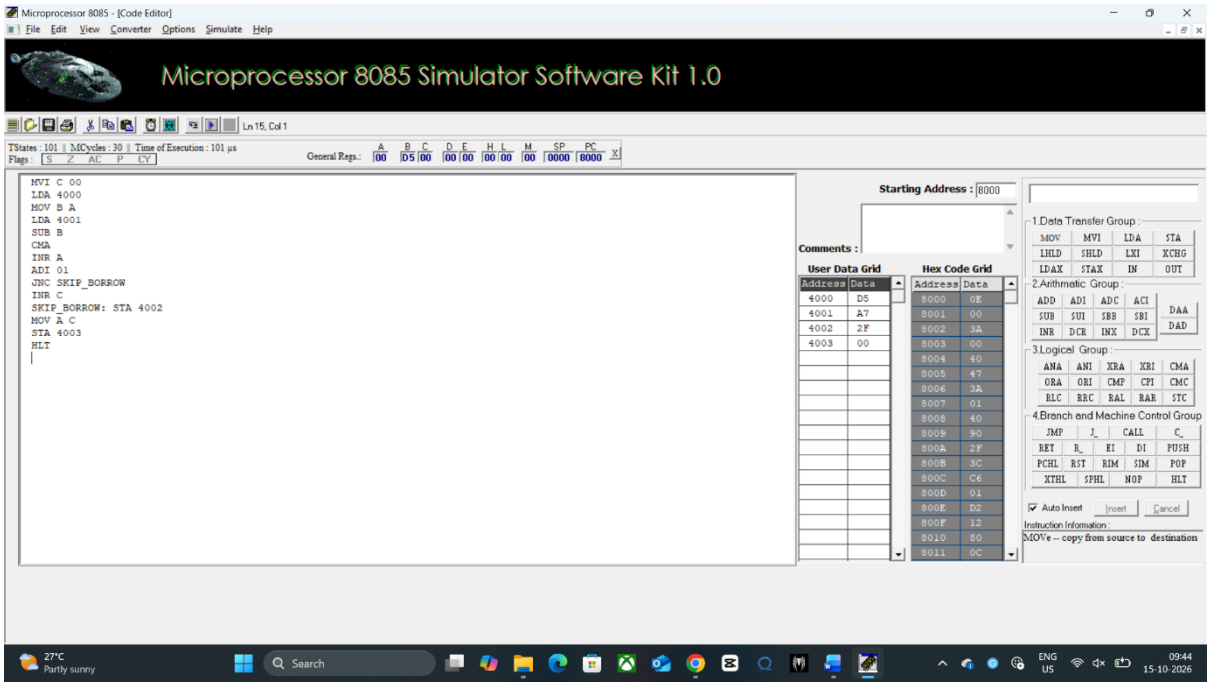


Data Table:

4000	**D5**
4001	**A7**
4002	**7C**
4003	**01**

Subtraction:

```
LDA 4000
MOV B A
LDA 4001
MVI C 00
SUB B
JNC SKIP
  INR C
  CMA
  ADI 01
SKIP: STA 4002
  MOV A C
  STA 4003
  HLT
```



Data Table:

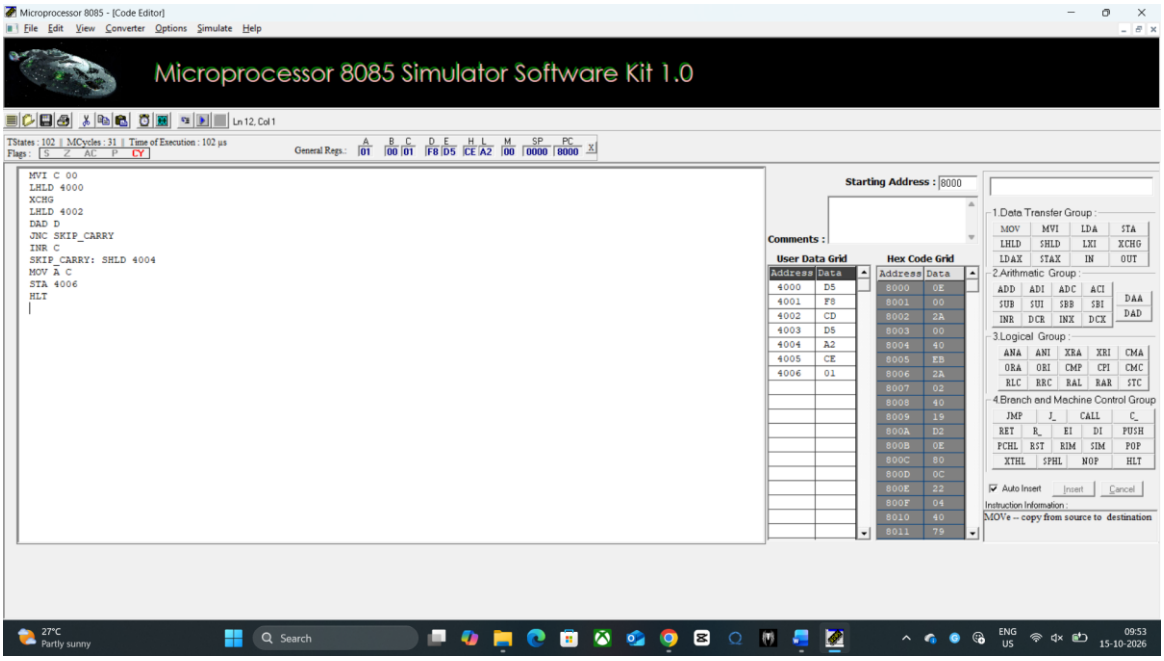
4000	**D5**
4001	**A7**
4002	**7C**
4003	**01**

2. **To Analysis and Verify 16 bit Addition and Subtraction using 8 bit Micro Processor**

Code:

Addition:

```
MVI C 00
LHLD 4000
XCHG
LHLD 4002
DAD D
JNC SKIP_CARRY
INR C
SKIP_CARRY: SHLD 4004
MOV A C
STA 4006
HLT
```

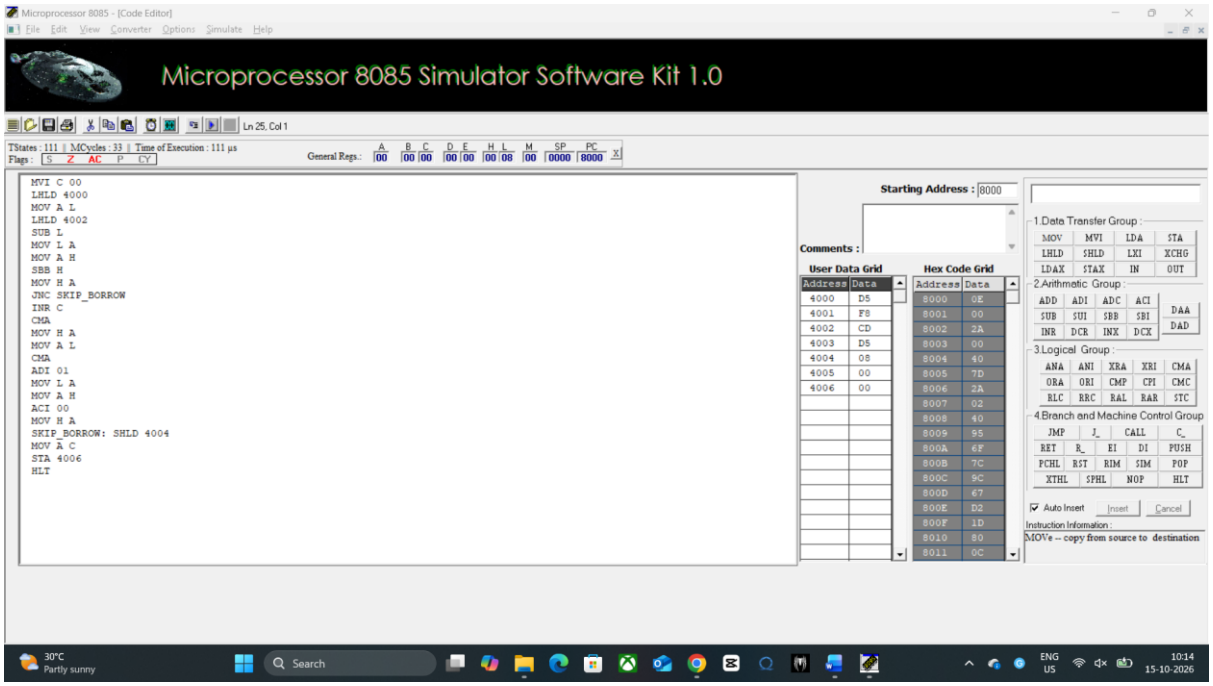


Data Table:

4000	**D5**
4001	**F8**
4002	**CD**
4003	**D5**
4004	**A2**
4005	**CD**
4006	**01**

Subtraction:

```
MVI C 00
LHLD 4000
MOV A L
LHLD 4002
SUB L
MOV L A
MOV A H
SBB H
MOV H A
JNC SKIP_BORROW
INR C
CMA
MOV H A
MOV A L
CMA
ADI 01
MOV L A
MOV A H
ACI 00
MOV H A
SKIP_BORROW: SHLD 4004
MOV A C
STA 4006
HLT
```



Data Table:

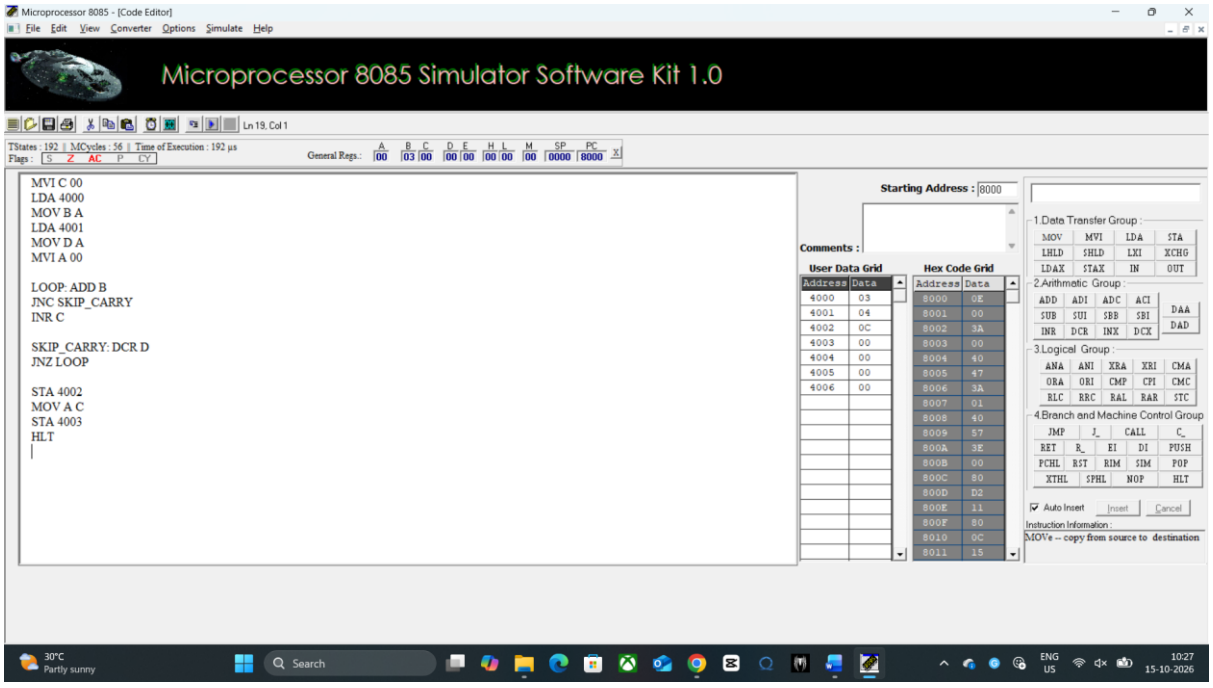
4000	**D5**
4001	**F8**
4002	**CD**
4003	**D5**
4004	**08**
4005	**23**
4006	**00**

3. **To Analysis and Verify 8 bit Multiplication and Division using 8 bit Micro Processor**

Code:

Multiplication:

```
MVI C 00
LDA 4000
MOV B A
LDA 4001
MOV D A
MVI A 00
LOOP: ADD B
JNC SKIP_CARRY
INR C
SKIP_CARRY: DCR D
JNZ LOOP
STA 4002
MOV A C
STA 4003
HLT
```



Data Table:

4000	**03**
4001	**04**
4002	**0C**
4003	**00**

Division:

MVI C 00

LDA 4000

LDA 4001

MOV B A

LDA 4000

LOOP: CMP B

JC STORE

SUB B

INR C

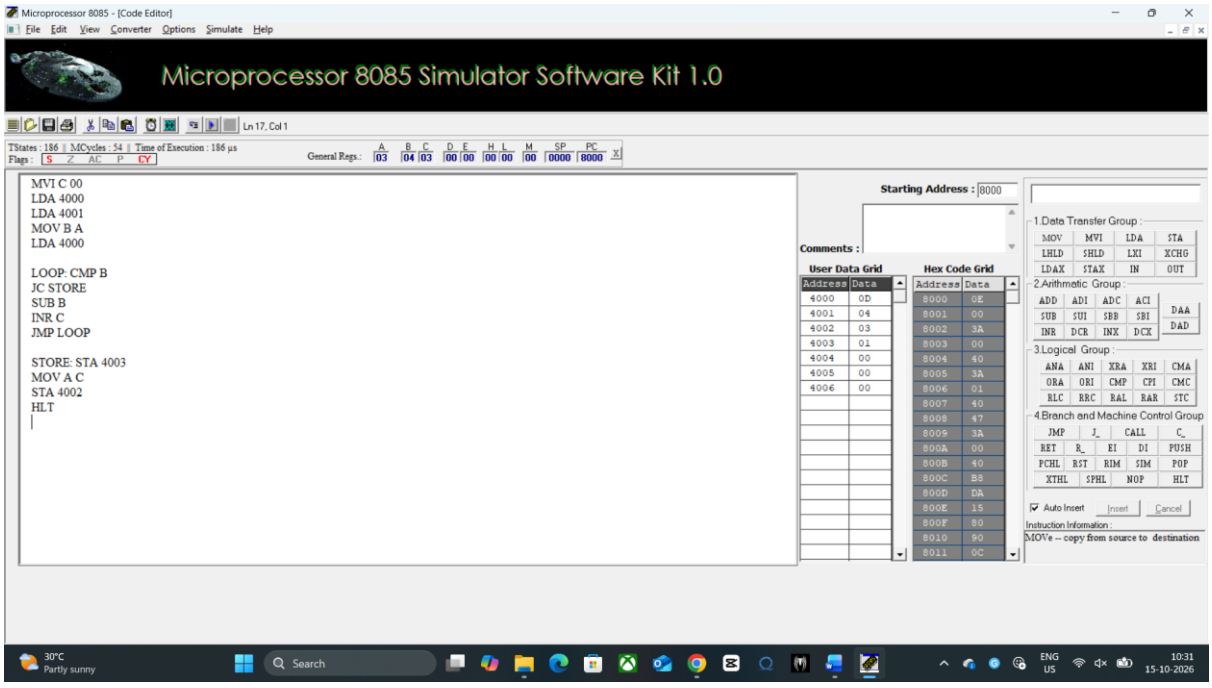
JMP LOOP

STORE: STA 4003

MOV A C

STA 4002

HLT



Data Table:

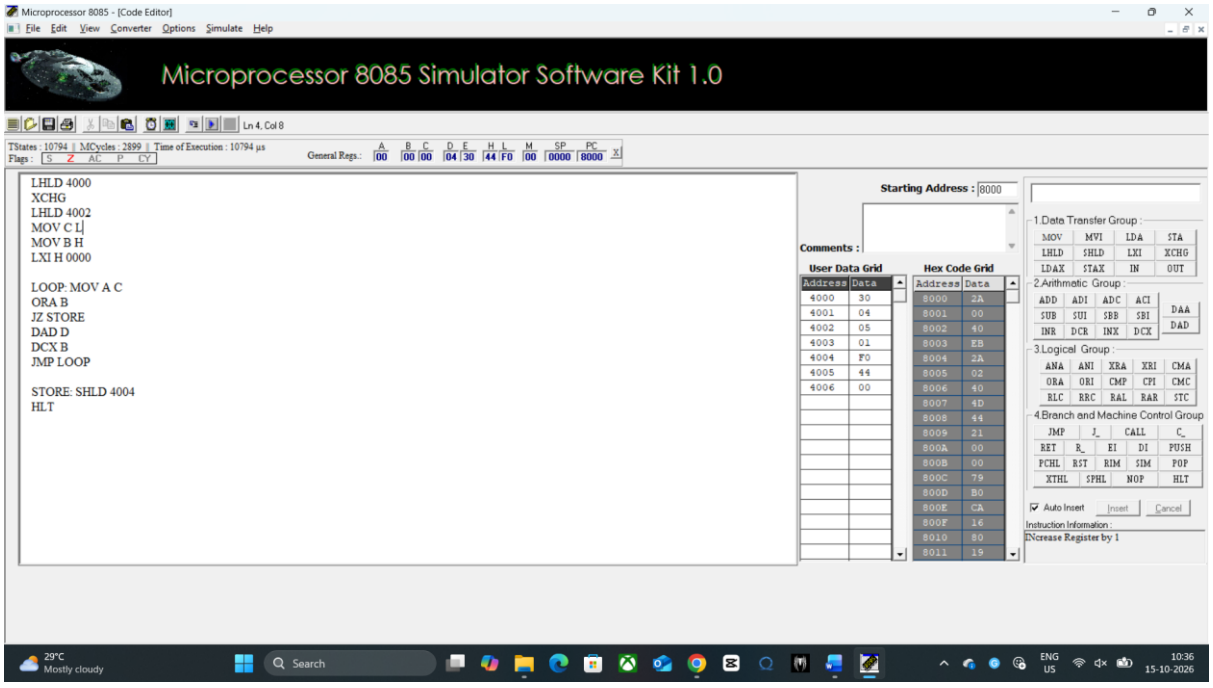
4000	**0D**
4001	**04**
4002	**03**
4003	**01**

4. **To Analysis and Verify 16 bit Multiplication and Division using 8 bit Micro Processor**

Code:

Multiplication:

```
LHLD 4000
XCHG
LHLD 4002
MOV C L
MOV B H
LXI H 0000
LOOP: MOV A C
ORA B
JZ STORE
DAD D
DCX B
JMP LOOP
STORE: SHLD 4004
HLT
```



Data Table:

4000	**03**
4001	**04**
4002	**05**
4003	**01**
4004	**F0**
4005	**44**
4006	**00**

Division:

LXI B 0000

LHLD 4000

XCHG

LHLD 4002

LOOP: MOV A E

SUB L

MOV A D

SBB H

JC STORE

MOV A E

SUB L

MOV E A

MOV A D

SBB H

MOV D A

INX B

JMP LOOP

STORE: XCHG

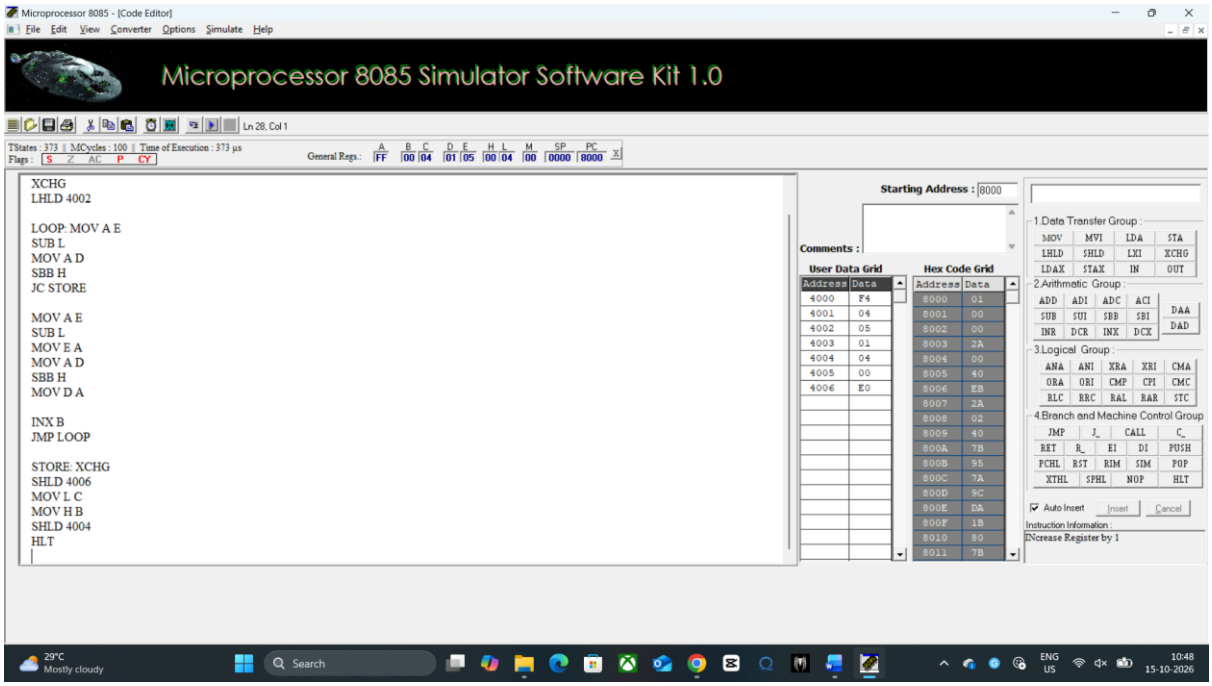
SHLD 4006

MOV L C

MOV H B

SHLD 4004

HLT



Data Table:

4000	**F4**
4001	**3D**
4002	**05**
4003	**01**
4004	**3C**
4005	**00**
4006	**C8**

Result:

1. 8-bit Addition and Subtraction: The target memory location holds the correct 8-bit sum or difference, and the adjacent byte records 01H if an overflow carry or borrow occurred, or 00H if it did not.
2. 16-bit Addition and Subtraction: The sequential memory addresses store the low and high bytes of the correct 16-bit sum or difference, along with a final byte reflecting the state of the terminal carry or borrow flag.
3. 8-bit Multiplication and Division: The output registers or memory addresses contain either a combined 16-bit product from repeated addition, or two distinct 8-bit bytes representing the exact Quotient and Remainder from repeated subtraction.
4. 16-bit Multiplication and Division: The designated target memory blocks accurately hold a multi-byte product (up to 32 bits) generated by the double-addition loop, or a pair of 16-bit values containing the final Quotient and Remainder.